INTRODUCTION TO ATTENTION & MONITORING

ENP-0112

2/9/2026

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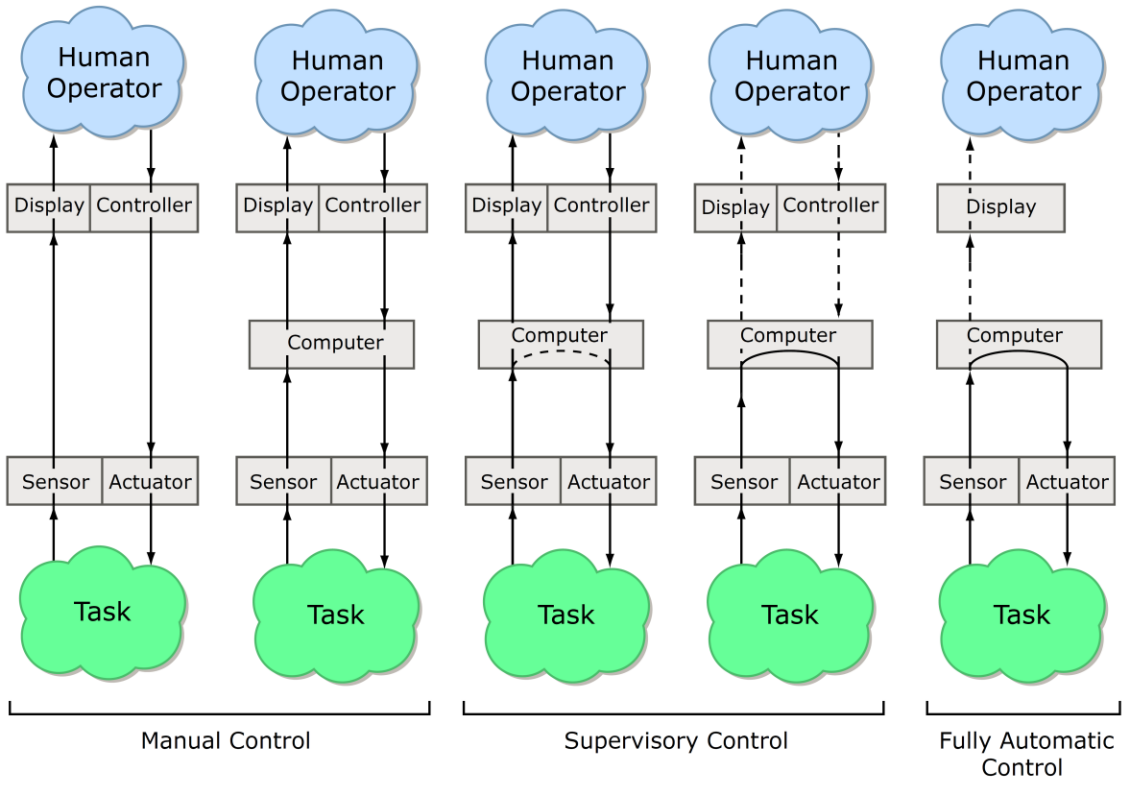
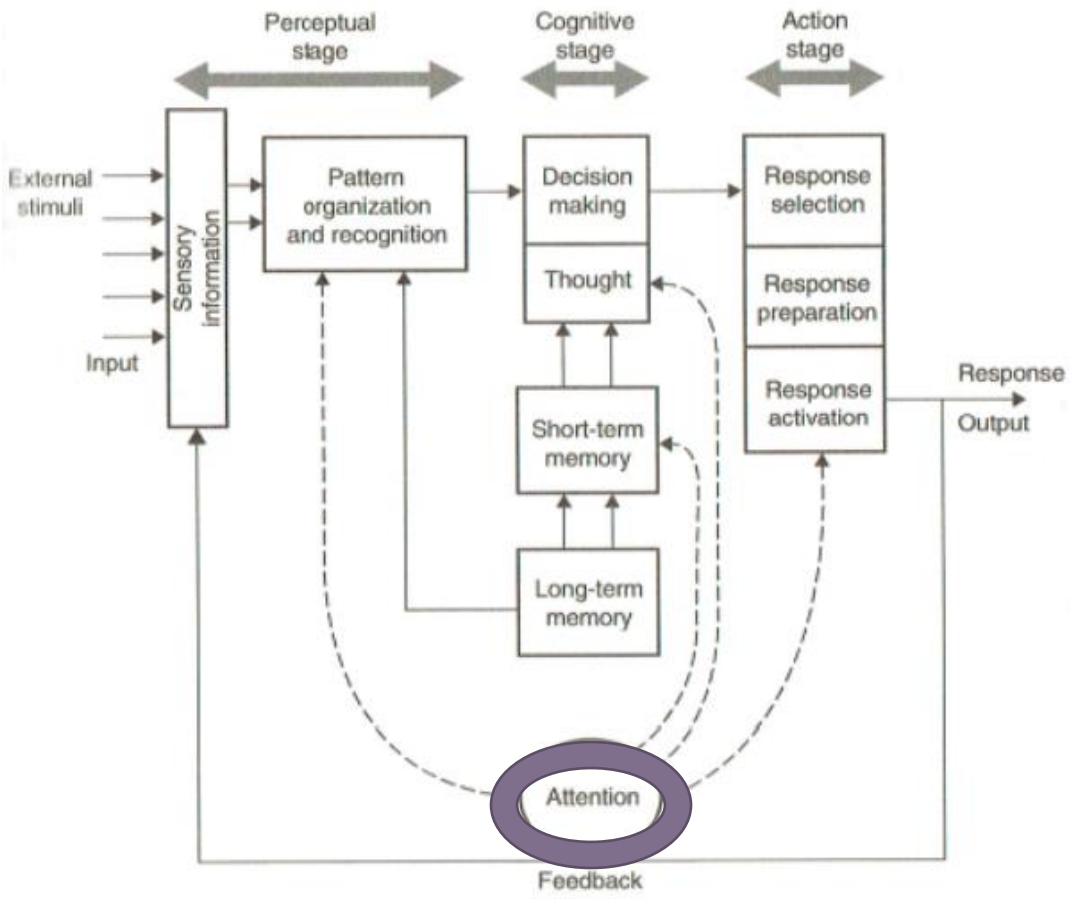
TOPICS

- Supervisory control of automation
- Attention and vigilance
- Monitoring techniques
- Discussion of flight path monitoring reading
- Extras
  - PAPI video
  - Models of attention

SHERIDAN'S SPECTRUM OF AUTOMATION



HUMAN INFORMATION PROCESSING MODEL

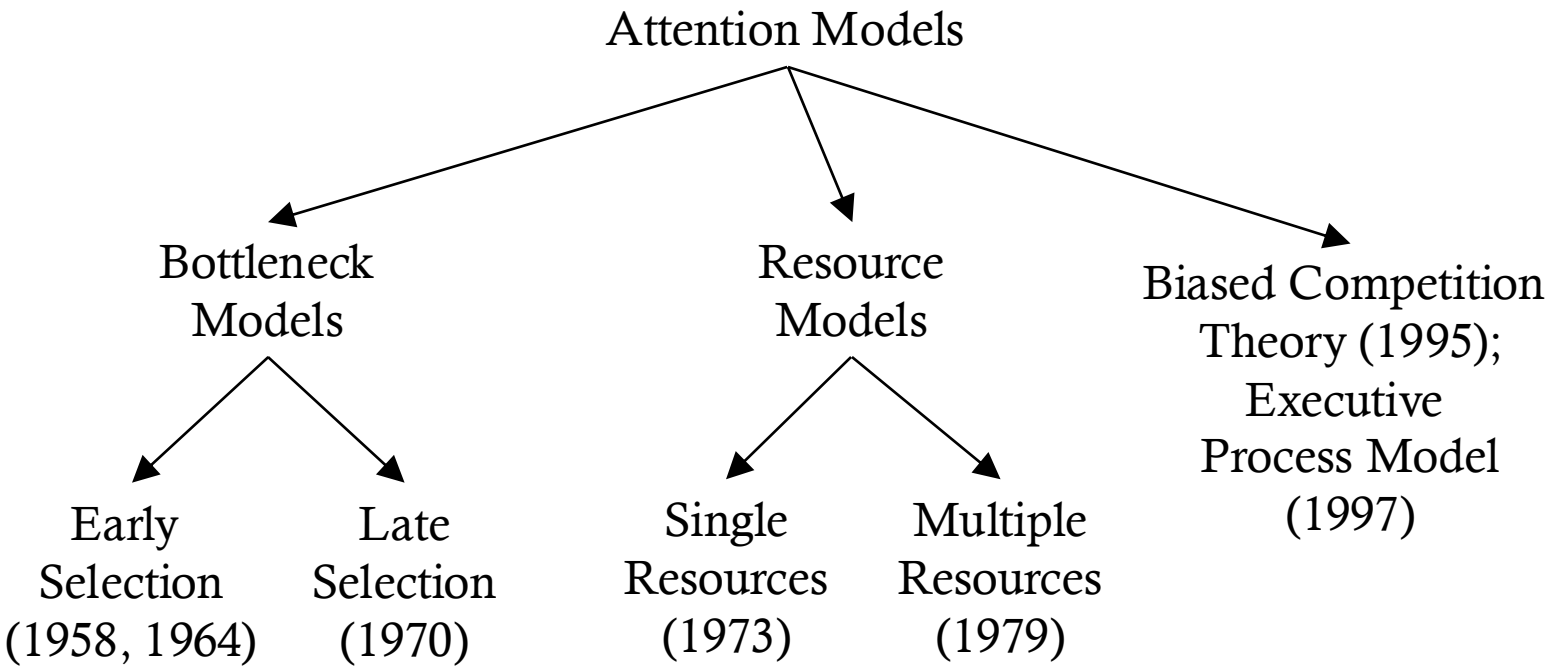


Proctor and VanZandt, 2017

WHAT IS ATTENTION?

- Attention is associated with “consciousness” and “awareness”
- Attention “connects” perception and cognition
  - It is an “executive” function
- We use metaphors to model attention
  - Spotlight, filter, bottleneck, (limited) resource
- Some words to describe attention
  - Focused, selective, divided, distributed

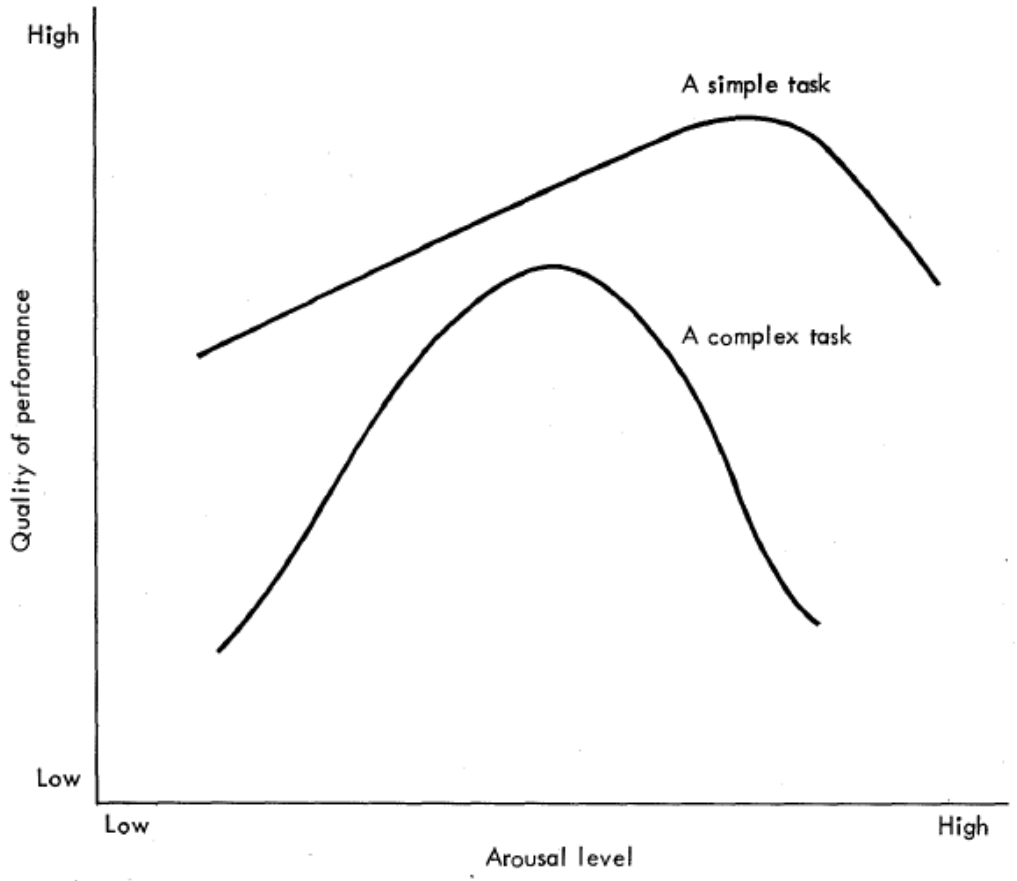
OVERVIEW: MODELS OF ATTENTION



See Proctor & Van Zandt, 2017

YERKES-DODSON LAW

An observed relationship between arousal and performance

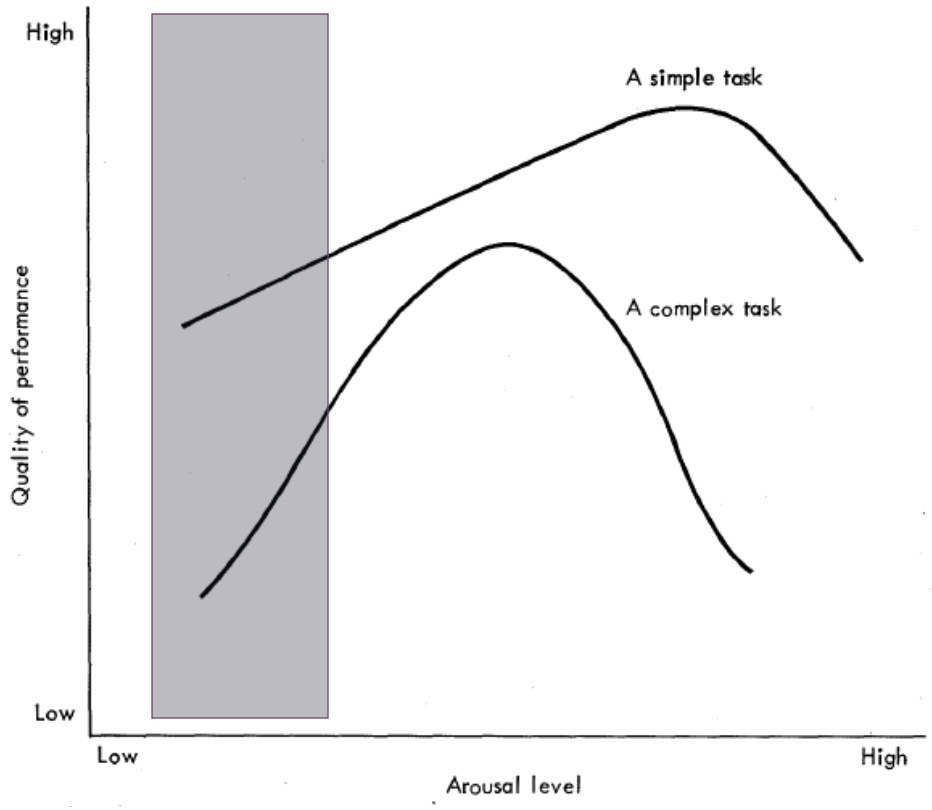


Kahneman D. Attention and Effort. Prentice Hall, Inc., 1973.

VIGILANCE

Vigilance is sustained attention, the ability to maintain attention/arousal. (Vigilance is hard.)

- Factors
  - Time
  - Salience
  - Signal rate
  - Arousal level

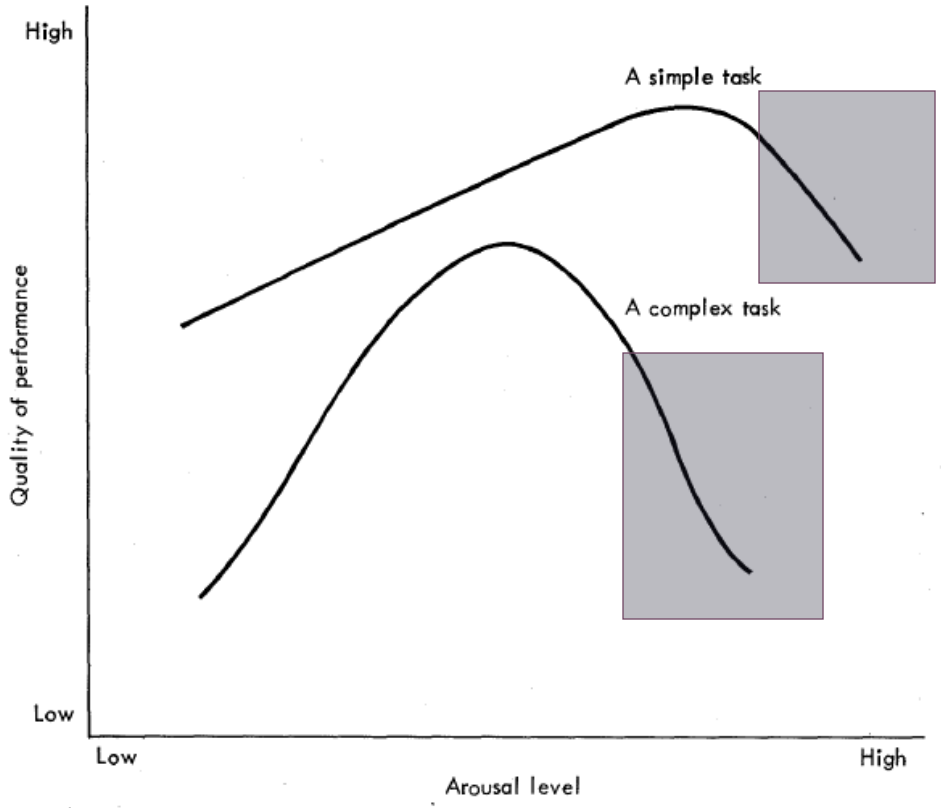


Monitoring is a vigilance task.

PERCEPTUAL NARROWING

Restricted attention due to high demands

- Increased tendency to focus on a few cues
- Selection of relevant cues often involves a discrimination
- High workload impairs discrimination between relevant and non-relevant cues
- Reduced ability to focus on relevant cues



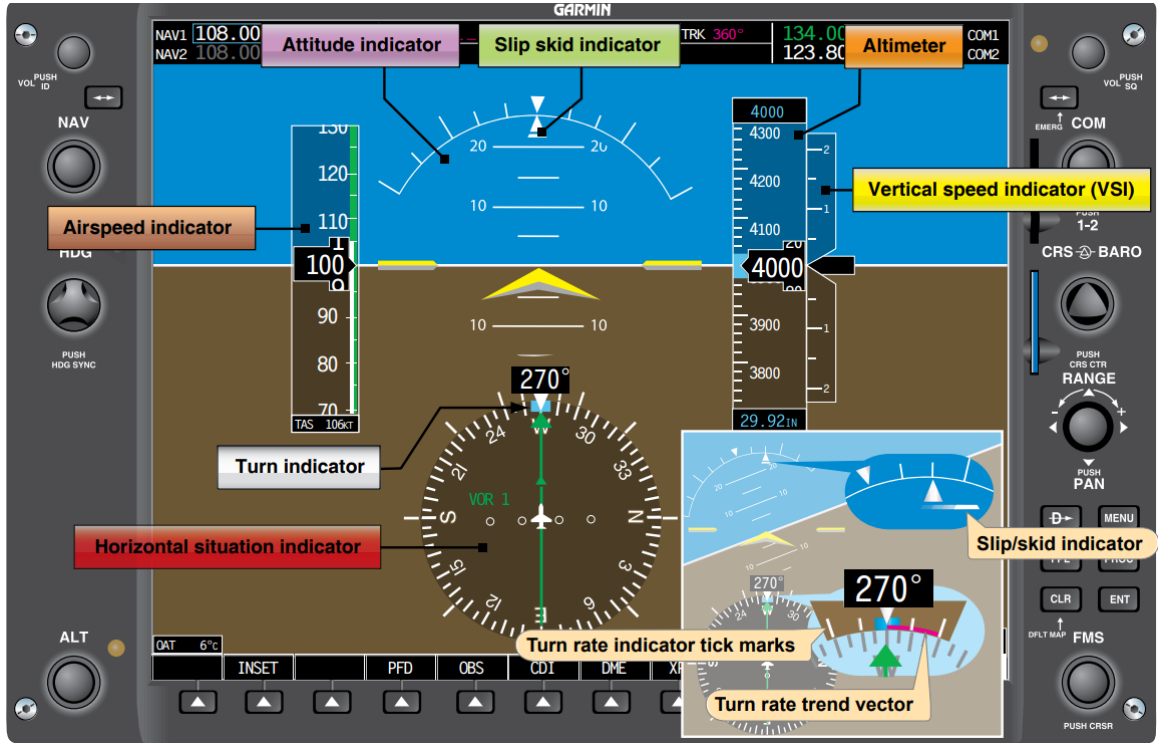
SOME MONITORING TECHNIQUES

- Scanning
  - Visual scan for traffic
  - Instrument scan
  - Cross checking (independent sources)
- Flows/checklists
- Point and Call (double-pointing)

INSTRUMENT SCAN

Basic T

Requires practice and proficiency!



FAA Pilot's Handbook of Aeronautical Knowledge, FAA-H-8083-25B

?

https://commons.wikimedia.org/wiki/File:Six_flight_instruments.JPG

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HEAD-UP DISPLAY

- Collimated display (at infinity)
- Integrated data
- Attention is a different issue…

http://aviationknowledge.wikidot.com/aviation:human-factors-issue-associated-with-heads-up-displa

MONITORING AND SENSE-MAKING

- Mental Models and Situation Awareness, covered later in class

Monitoring, essentially, is "sense making" (Section 3.1). It is systematic observation and interpretation of the current state of the airplane and its operational environment; it requires integration of current inputs with operational knowledge, which includes mental models, and the generation of expected values of flight path targets.

Monitoring is tied to expectations (Section 3.1). Monitoring—looking and listening to gather information about the operational environment—is not simply moving one's eyes to each indication and reading it. Typically, but not always, there is an expectation about what that indication will be, perhaps based on a projection from the last glance.

Mumaw, Billman, & Feary (2020) Analysis of pilot monitoring skills and a review of training effectiveness. (NASA/TM-20210000047) (in Canvas, under Supplementary readings)

HAND SIGNALS FOR MONITORING

- Japanese hand rail signals (4:16)

DISCUSSION: PRACTICAL GUIDE TO FLIGHT PATH MONITORING

- Do you have personal experience in monitoring?
  - For what purpose?
  - What are your reflections about the monitoring task?
- How might (or might not) this apply to other modes of transportation?

Flight Safety Foundation (November, 2014). *A practical guide for improving flight path monitoring.*

SOME REFERENCES

Proctor and Van Zandt textbook (in Canvas under Reading List)

Flight Safety Foundation (2014). *A practical guide for improving flight path monitoring.*

Mumaw, Billman, & Feary (2020) Analysis of pilot monitoring skills and a review of training effectiveness. (NASA/TM-20210000047) (in Canvas, under Supplementary readings)

EXTRAS

APPROACH VIDEOS
LED VS. INCANDESCENT PAPI

- PAPI/VASI video

SPOTLIGHT METAPHOR

- Attention can be aligned with direction of gaze, or directed to be independent of fixation
- What you are “attending to” can be different from where your eyes are focused (e.g., team sports)

Abstract Colored Circles Linked with Coloured Bands by Paul Klee 1914

BOTTLENECK (FILTER) MODELS OF ATTENTION

- Selective attention tasks (typically auditory)

- Where exactly is the bottle neck (early or late) in information processing?
  - Early (Broadbent's filter theory)
  - Late (Treisman's filter attenuation theory, Deutsch & Deutsch)

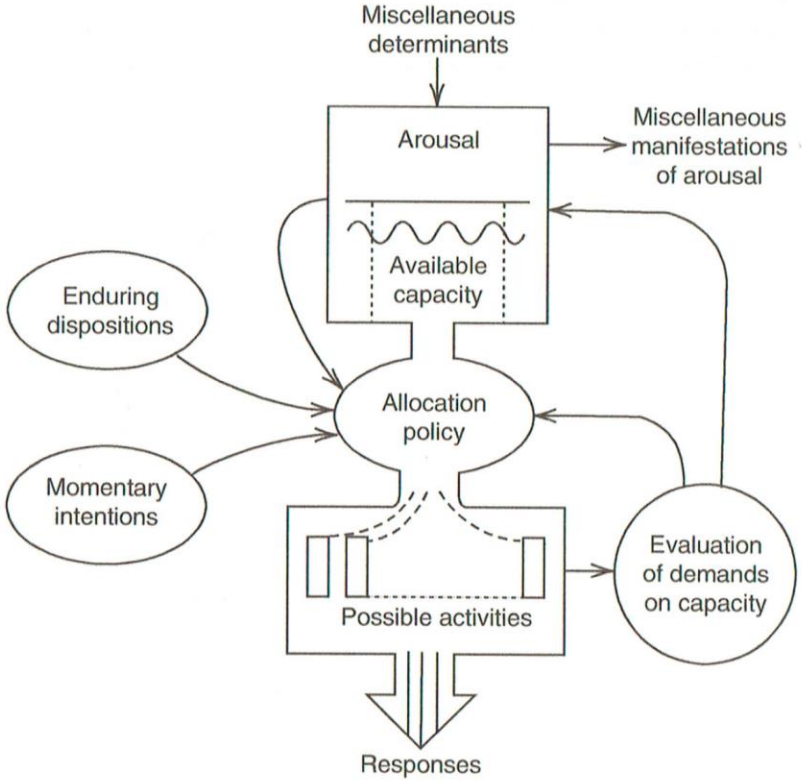
- Is the bottleneck a structural limitation (bug) or an optional strategy (feature)?

MORE RECENT MODELS OF ATTENTION

- Spatially-based Feature-Integration Theory (Treisman)
- Attention as a unitary resource (e.g., Kahneman)
- Attention as multiple resources (Wickens)
- Executive Control Model (EPIC) (Meyer & Kieras)

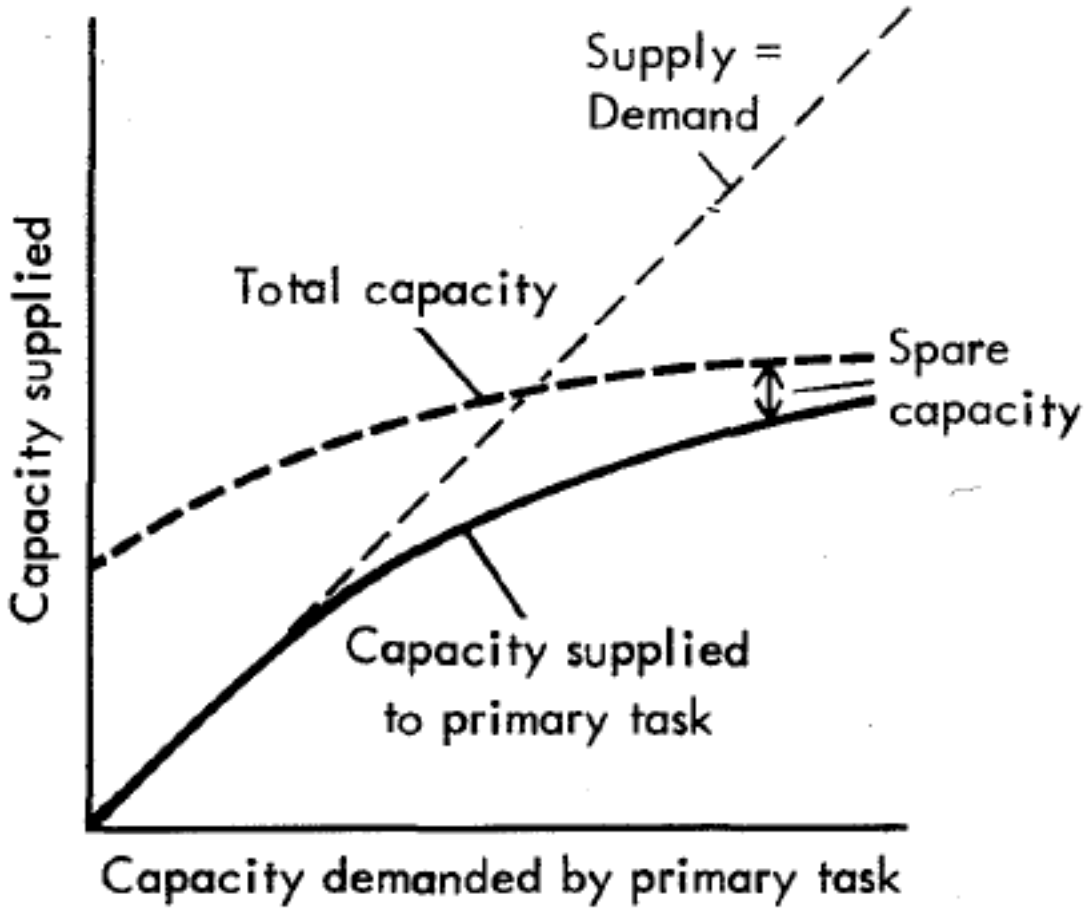
RESOURCE MODELS – SINGLE RESOURCES

Attention is a resource that can be divided up among different tasks in different amounts



Kahneman, 1973

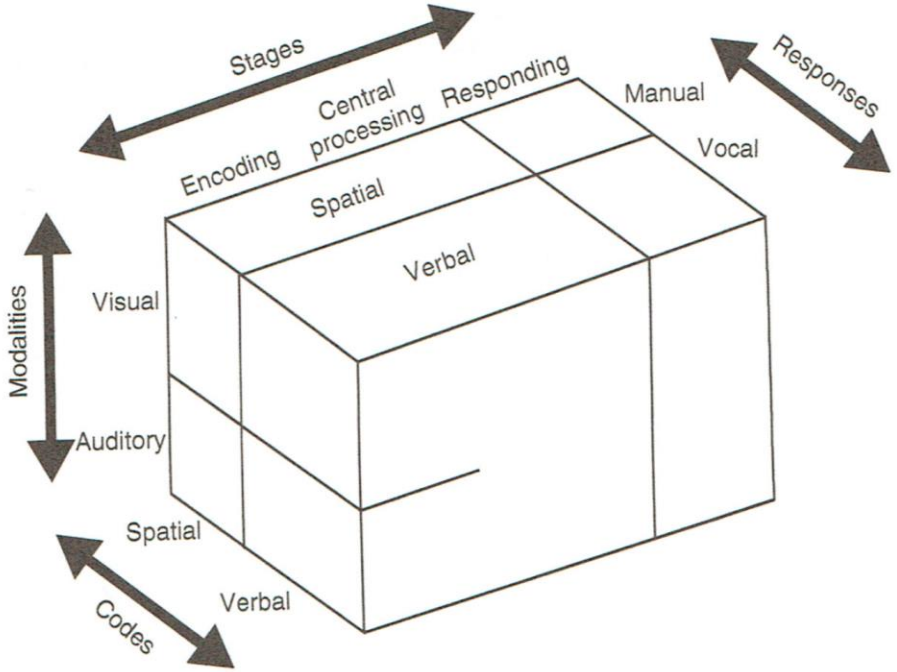
RESOURCE MODELS – SINGLE RESOURCES



Kahneman, 1973

RESOURCE MODELS – MULTIPLE RESOURCES

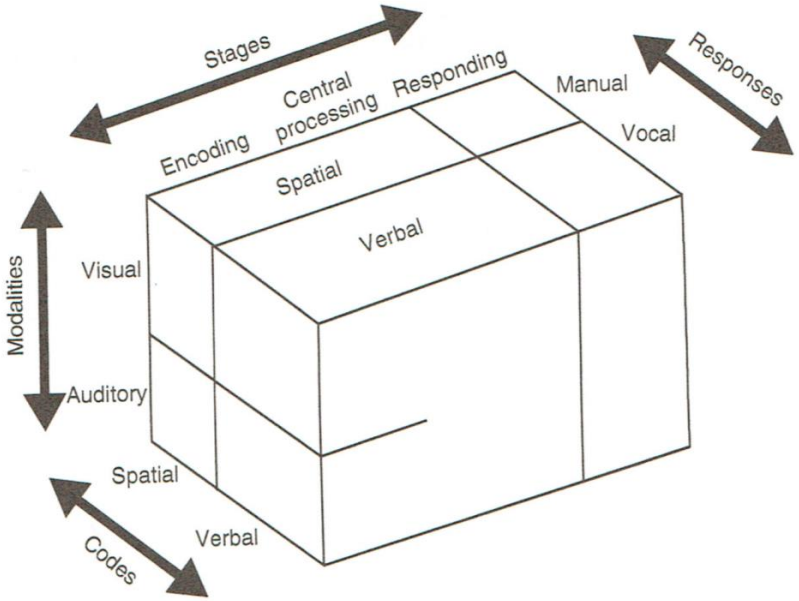
Different attentional resources exist for different sensory-motor modalities and coding domains



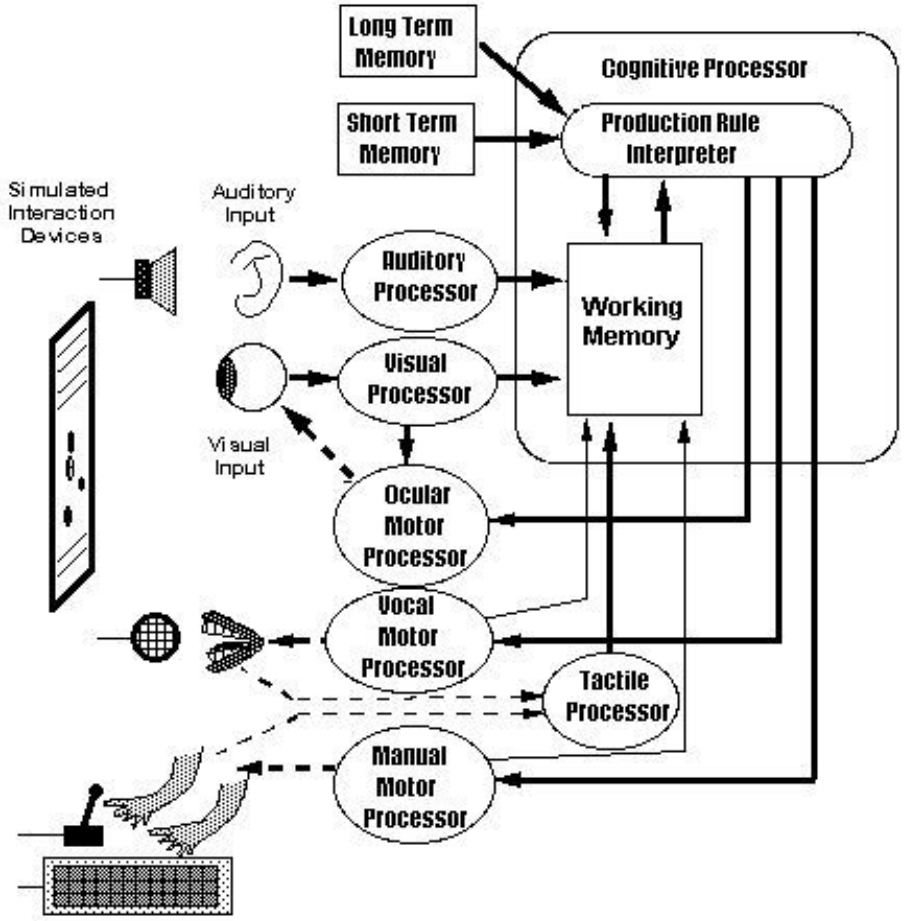
Proctor and Van Zandt, 2017

THOUGHT EXERCISE

Consider walking with a friend describing something to you…



EXECUTIVE CONTROL MODELS



- EPIC (Executive-Process/Interactive Control)

Meyer and Kieras, 1997

CHANGE BLINDNESS

- The inability to detect changes in a visual scene
- Focused attention is necessary to see change
- We do not attend to all visual details in our visual buffer (iconic memory)
- Change blindness occurs when attention is distracted or there is a break in visibility of information

Asking for Directions

What are some practical situations where this might occur, e.g., in driving or flying?